kvpara: Klein-Vella Parametric Control Function Estimation

Stata Implementation Documentation

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Abstract

This document provides a comprehensive guide to the estimation procedure implemented in the `kvpara` Stata command. The method follows Klein and Vella (2010) and Farré, Klein, and Vella (2013). Unlike instrumental variable approaches, identification comes from heteroskedasticity in the error terms, allowing estimation without exclusion restrictions.

Contents

1 Introduction and Key Distinction from IV

The `kvpara` command implements the Klein-Vella (2010) parametric control function approach for estimating models with endogenous regressors. This method is fundamentally different from traditional instrumental variable (IV) approaches.

1.1 Key Distinction: No Exclusion Restrictions Required

Unlike IV methods (2SLS, GMM), the Klein-Vella approach does **not require exclusion restrictions**. That is, you do not need instruments that affect the endogenous variable but are excluded from the main equation. Instead, identification comes from **heteroskedasticity** in the first-stage errors.

In practice, this means `controls_main()` and `controls_endog1()` typically contain the **same variables** (or `controls_endog1()` is a subset of `controls_main()`). This is intentional and correct—it is what distinguishes this approach from IV estimation.

Table 1: Comparison: IV vs Klein-Vella Approaches

Feature	**IV/2SLS**	**Klein-Vella**
Exclusion restrictions	Required	Not required
Identification source	Instrument relevance	Heteroskedasticity
`controls_main` vs `controls_endog`	Must differ	Can be identical
Key assumption	Instrument exogeneity	Correct het specification

2 The Econometric Framework

2.1 The Triangular System

Consider a triangular simultaneous equations model:

Main Equation (Wage/Outcome):

$$y = X'\beta + S \cdot \gamma + u \tag{1}$$

First Stage (Education/Treatment):

$$S = X'\pi + v \tag{2}$$

where:

- y is the outcome variable (e.g., log wage)
- S is the endogenous variable (e.g., years of education)
- X are control variables (the **same** in both equations—no exclusion restriction)
- γ is the parameter of interest (e.g., return to education)
- u and v are error terms that may be correlated

Important: Same Controls

Note that X appears in **both** equations. This is the key distinction from IV—we do not need variables that are in the first stage but excluded from the main equation.

2.2 The Endogeneity Problem

If $\text{Cov}(u, v) \neq 0$, then OLS estimation of γ is biased. The Klein-Vella method corrects for this by including a control function that captures the correlation between u and v.

2.3 Heteroskedasticity Structure

The key identifying assumption is that the error variances depend on observable covariates. Let Z denote the variables in the heteroskedasticity functions (typically a subset of X):

Main equation heteroskedasticity:

$$\text{Var}(u|Z) = \sigma_u^2(Z) = \exp(Z'\theta_u) \tag{3}$$

First stage heteroskedasticity:

$$\text{Var}(v|Z) = \sigma_v^2(Z) = \exp(Z'\theta_v) \tag{4}$$

The heteroskedasticity provides the variation needed for identification without requiring traditional instruments.

Notation Summary

- X = control variables in main and first-stage equations (`controls_main`, `controls_endog1`)
- Z = variables in heteroskedasticity functions (`het_main`, `het_endog1`), typically $Z \subseteq X$
- θ_u = heteroskedasticity parameters for main equation
- θ_v = heteroskedasticity parameters for first stage

3 Step-by-Step Estimation Procedure

The `kvpara` command implements the following four-step procedure:

3.1 Step 1: OLS Estimation (Baseline)

Run ordinary least squares on the main equation ignoring endogeneity:

$$y = X'\beta_{\text{OLS}} + S \cdot \gamma_{\text{OLS}} + \text{residual} \tag{5}$$

This provides a baseline for comparison. Heteroskedasticity tests (Breusch-Pagan and White) are conducted on the OLS residuals.

```
Stored results: e(b_ols), e(se_ols), e(bp_chi2_ols), e(
    white_chi2_ols)
```

3.2 Step 2: First Stage Estimation

Step 2a. Estimate the first stage equation:

$$S = X'\hat{\pi} + \hat{v} \tag{6}$$

Obtain residuals $\hat{v} = S - X'\hat{\pi}$. Heteroskedasticity tests are conducted.

Step 2b. Model heteroskedasticity in the first stage:

$$\log(\hat{v}^2) = Z'\theta_v + \text{error} \tag{7}$$

Step 2c. Obtain predicted standard deviation:

$$\hat{\sigma}_v = \sqrt{\exp(Z'\hat{\theta}_v)} \tag{8}$$

Note

These quantities ($\hat{v}$ and $\hat{\sigma}_v$) are computed once in Step 2 and remain **fixed** throughout the iterative procedure in Step 3.

```
Stored results: e(b_first1), e(se_first1), e(theta_he1), e(se_he1)
```

3.3 Step 3: Iterative Klein-Vella Estimation

This is the core of the procedure. The algorithm iterates between estimating the main equation and updating the heteroskedasticity model for the main equation.

Iteration 1: Pure OLS (no control function)

$$y = X'\beta + S \cdot \gamma + \text{residual} \tag{9}$$

Then model heteroskedasticity:

$$\log(\text{residual}^2) = Z'\theta_u \tag{10}$$

Compute $\hat{\sigma}_u = \sqrt{\exp(Z'\hat{\theta}_u)}$

Iterations 2 to T: Include control function

$$y = X'\beta + S \cdot \gamma + \text{CF} \cdot \rho + \text{residual} \tag{11}$$

where the control function is:

$$\boxed{\text{CF} = \frac{\hat{\sigma}_u}{\hat{\sigma}_v} \times \hat{v} = \text{scale} \times \hat{v}} \tag{12}$$

After each regression:

1. Update the heteroskedasticity model: $\log(\text{residual}^2) = Z'\theta_u$
2. Recompute $\hat{\sigma}_u = \sqrt{\exp(Z'\hat{\theta}_u)}$
3. Update the control function: $\text{CF} = (\hat{\sigma}_u/\hat{\sigma}_v) \times \hat{v}$

The procedure typically converges within 10 iterations (the default).

```
Stored results: e(b_kv), e(se_kv), e(theta_hmain), e(se_hmain)
```

3.4 Step 4: Bootstrap Standard Errors (Optional)

Bootstrap standard errors are recommended because they properly account for estimation uncertainty in:

- First stage coefficient estimation
- Heteroskedasticity function estimation
- Control function construction

Each bootstrap replication re-estimates the entire procedure (Steps 1–3) on a resampled dataset. Standard errors are computed from the bootstrap distribution.

4 The Control Function and Its Components

The control function corrects for endogeneity by capturing the correlation between the first-stage residual and the main equation error. It consists of three components:

$$\text{CF} = \underbrace{\frac{\hat{\sigma}_u}{\hat{\sigma}_v}}_{\text{scale}} \times \underbrace{\hat{v}}_{\text{first-stage residual}} \tag{13}$$

Table 2: Control Function Components

Component	Variable Name	Description	Source
$\hat{\sigma}_u$	`stub_su`	Predicted SD from main eq het function	Final iteration
$\hat{\sigma}_v$	`stub_sv1`	Predicted SD from first stage het function	Step 2 (fixed)
$\hat{v}$	`stub_vhat1`	First stage residual	Step 2 (fixed)
scale	`stub_scale1`	Ratio $\hat{\sigma}_u/\hat{\sigma}_v$	Computed
CF	`stub_cf1`	Control function = scale $\times$ $\hat{v}$	Computed

4.1 Generating CF Components

Use the `generate(stub, replace)` option to create variables containing these components:

```
kvpara lwage educ, controls_main(X) controls_endog1(X) ///
    het_main(Z) het_endog1(Z) generate(kv, replace)
```

4.2 Summary Statistics in e()

Summary statistics for CF components are always stored in `e()`, regardless of whether `generate()` is specified:

Scalar	Description
`e(mean_su)`	Sample mean of $\hat{\sigma}_u$
`e(sd_su)`	Sample SD of $\hat{\sigma}_u$
`e(mean_sv1)`	Sample mean of $\hat{\sigma}_v$
`e(sd_sv1)`	Sample SD of $\hat{\sigma}_v$
`e(mean_scale1)`	Sample mean of scale ($\hat{\sigma}_u/\hat{\sigma}_v$)
`e(sd_scale1)`	Sample SD of scale
`e(mean_cf1)`	Sample mean of CF
`e(sd_cf1)`	Sample SD of CF

5 Stored Results Reference

5.1 Scalars

5.2 Matrices

6 Practical Usage Examples

6.1 Basic Estimation

Scalar	Description
e(N)	Number of observations
e(nendog)	Number of endogenous variables
e(iterations)	Number of iterations
e(bootstrap)	Number of bootstrap replications
e(boot_success)	Number of successful bootstrap replications
e(r2_first1)	R-squared for first stage
e(F_first1)	F-statistic for first stage
e(bp_chi2_first1)	Breusch-Pagan χ^2 for first stage
e(bp_p_first1)	Breusch-Pagan p-value for first stage
e(white_chi2_first1)	White test χ^2 for first stage
e(white_p_first1)	White test p-value for first stage
e(bp_chi2_ols)	Breusch-Pagan χ^2 for OLS
e(white_chi2_ols)	White test χ^2 for OLS

Matrix	Description
e(b)	Klein-Vella coefficient vector (posted)
e(V)	Klein-Vella variance-covariance matrix (posted)
e(b_ols)	OLS coefficient vector
e(se_ols)	OLS standard error vector
e(b_kv)	Klein-Vella coefficient vector
e(se_kv)	Klein-Vella standard error vector
e(b_first1)	First stage coefficient vector
e(se_first1)	First stage standard error vector
e(theta_hmain)	Main equation het function coefficients
e(se_hmain)	Main equation het function SEs
e(theta_he1)	First stage het function coefficients
e(se_he1)	First stage het function SEs

```
* Define controls (same for main and first stage)
local X age age_sq female region_d1 region_d2 region_d3

* Define het variables (subset of X)
local Z age age_sq female

* Estimate Klein-Vella model
kvpara lwage educ, ///
    controls_main(‘X’) ///
    controls_endog1(‘X’) ///
    het_main(‘Z’) ///
    het_endog1(‘Z’) ///
    iterations(10) bootstrap(100)
```

6.2 Accessing Key Results

```
* Klein-Vella coefficients and standard errors
matrix list e(b_kv)
matrix list e(se_kv)

* Return to education coefficient
display "Return to education: " _b[educ]
display "Standard error: " _se[educ]
```

```
* Control function coefficient (endogeneity test)
display "CF coefficient: " _b[_cor1]
display "CF t-stat: " _b[_cor1]/_se[_cor1]

* Heteroskedasticity function (main equation)
matrix list e(theta_hmain)

* First stage heteroskedasticity test
display "BP test p-value: " e(bp_p_first1)

* CF component summary statistics
display "Mean sigma_u: " e(mean_su)
display "Mean sigma_v: " e(mean_sv1)
display "Mean scale: " e(mean_scale1)
display "Mean CF: " e(mean_cf1)
```

6.3 Generating CF Components

```
kvpara lwage educ, ///
    controls_main(`X') controls_endog1(`X') ///
    het_main(`Z') het_endog1(`Z') ///
    bootstrap(100) generate(kv, replace)

* Verify the identity: CF = scale * V-hat
gen check = kv_scale1 * kv_vhat1
assert abs(check - kv_cf1) < 1e-10
drop check

* Summarize components
summarize kv_su kv_sv1 kv_scale1 kv_cf1
```

6.4 Adding CF Statistics to Tables

```
* After kvpara estimation, add formatted statistics
estadd local mean_su_str = string(e(mean_su), "%6.3f") ///
    + " (" + string(e(sd_su), "%6.3f") + ")"
estadd local mean_sv_str = string(e(mean_sv1), "%6.3f") ///
    + " (" + string(e(sd_sv1), "%6.3f") + ")"
estadd local mean_scale_str = string(e(mean_scale1), "%6.3f") ///
    + " (" + string(e(sd_scale1), "%6.3f") + ")"
estadd local mean_cf_str = string(e(mean_cf1), "%6.3f") ///
    + " (" + string(e(sd_cf1), "%6.3f") + ")"
```

7 References

Klein, R. and F. Vella. 2010. Estimating a Class of Triangular Simultaneous Equations Models Without Exclusion Restrictions. *Journal of Econometrics* 154: 154–164.

Farré, L., Klein, R. and F. Vella. 2013. A Parametric Control Function Approach to Estimating the Returns to Schooling in the Absence of Exclusion Restrictions: An Application to the NLSY. *Empirical Economics* 44(1): 111–133.